

Qn01. Diabetes Data

1.1 Checking for correlation among explanatory variables

Observation

1. There is high positive correlation (0.90) between S1 and S2 variables, indicating strong multicollinearity that can destabilize regression coefficient estimates. It is recommended to either combine or remove one of these variables.
2. The remaining pairs of independent variables exhibit correlations below concerning thresholds, with the strongest alternative correlation being a negative relationship of -0.74 between S3 and S4, which warrants consideration but is less problematic than the S1-S2 relationship.

1.2 Collinearity

- Collinearity is a case when two or more predictor variables in a regression model are highly correlated, which means that they move together.
- Collinearity amongst predictor features can lead to estimated coefficient values becoming highly sensitive to random errors in the observed target, producing a large variance (Inflated Standard Errors)

1.3 Training a linear regression model

Observation

- Not all variables are significant as observed S6 and S3 are below the alpha level (0.05)
- The condition number is large, $7.24e+03$. This might indicate that there are strong multicollinearity or other numerical problems.

1.4 Forward & Backward Selection

Differences between forward and backward selection are:

➤ **Stepwise forward selection:**

- Start with empty set
- Add best attribute iteratively

WHILE

➤ **Stepwise backward elimination:**

- Start with full set
- Remove worst attribute iteratively

1.5 Forward Selection Stepwise Function

- The stepwise approach works by adding or removing predictor variables based on their statistical significance. The process operates by comparing different potential models using a pre-defined criterion, such as p-values, Akaike Information Criterion or Bayesian Information Criterion. The procedure stops when no further addition or removal of a variable significantly improves the model's performance according to the chosen criterion.
- The `forward_regression()` function works by starting with no variables and adds the one with the lowest p-value to the `best_features` array at each step, provided it is below a threshold (e.g., 0.05). It then returns the `best_features` array when no further addition can be done.

Qn02. Analyzing the Titanic Data set

2.1 Difference between linear and logistic regression

- The difference between linear regression and logistic regression is that logistic regression is a supervised learning classification algorithm that predicts the probability of a categorical output while linear regression is a supervised learning regression algorithm which predicts a continuous quantity

2.4 Parameter Statistical Significance

- The parameter estimates are `pclass` (-1.0691), `sex` (-2.4984) and `age` (-0.0318) also the parameters are statistically significant because their p-values are less than 0.05

Qn03. PCA

3.1 Definition and importance of PCA

Principal Component Analysis (PCA) is a method of performing dimensionality reduction on a dataset that helps in identifying and selecting features that can explain a large percent of the dataset, thus reducing the complexity of the dataset in machine learning.

It is useful to consider PCA to transform a set of explanatory variables because:

- It reduces dataset size
- It makes patterns easier to understand which prevents overfitting
- It speeds up ML model training
- It simplifies data for visualization

3.2 Mathematical equations for PCA

The mathematical equation for PCA is:

$$T = XW$$

whereby:

T (Score Matrix): An $n \times k$ matrix representing the data in the new coordinate system. Each row is a transformed observation, and each column is a principal component.

X (Centered Input Matrix): An $n \times p$ matrix where the mean of each column has been subtracted ($X = X_{raw} - \mu$).

W (Weight Matrix): A $p \times k$ matrix whose columns are the **eigenvectors** of the data's covariance matrix

3.3 DJI Principal Component Analysis

- The first principal component is similar to the market which suggests that the component is capturing the overall movement of the Dow Jones as a whole.